

Dhirubhai Ambani University
First In-Semester Examination, Autumn Semester 2025-2026

Course Title IT645 Web and Mobile Development
Date 21st Aug 2025

Max Marks 35
Time 1 Hour

Instructions:

1. All questions are compulsory.
2. Write your answers briefly; writing long answers does not fetch higher marks.
4. Make suitable assumptions wherever necessary.

1. Consider the code below and answer the following questions. (2+2 marks)

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
width: 100px;
height: 100px;
border: 5px solid black;
padding: 10px;
margin: 20px;
background-color: lightgreen;
}
</style>
</head>
<body>
<div>Box</div>
</body>
</html>
```

- a. Draw the CSS box model for the div element with exact dimensions.
 - b. Background color will be applied to which parts? Label in the above diagram.
2. Consider the following line in robots.txt: (2+2 marks)
- disallow: /private/
- a. What does this statement do?
 - b. Will it block users from accessing the folder /private/? Why or why not?
3. Consider the CSS below. (4 marks)

```
.card {
background-color: #ffffff;
padding: 24px;
border: 1px solid #0000ff;
border-radius: 10px;
}
```

- a. Write the equivalent Tailwind CSS classes to apply the same styles for a paragraph element.

4. Consider the code below and answer. (2+2 marks)

```
<head>
<style>
.size {
  width: 160px;
  height: 60px;
}
</style>
</head>

<body>
<section >
  <a href="#" class="size">Link</a>
  <span class="size">Span text</span>
  
  <button class="size">Button</button>
  <input class="size" value="Input field" />
  <textarea class="size">Textarea</textarea>
  <div class="size">Div content</div>
  <p class="size">Paragraph content</p>
  <ul class="size">
    <li>List item</li>
  </ul>
</section>
</body>
```

- For which of the elements in the <body>, CSS selector .size applies? Justify.
- Without changing the HTML, what single CSS rule would make <a> and start respecting width and height?

5. Consider the code below and answer. (2+2 marks)

```
<head>
<style>
html {
  font-size: 16px;
}
.box {
  width: 2rem;
  height: 50vh;
  background-color: lightblue;
  position: relative;
}
.box::after {
  content: "Demo";
  position: absolute;
  bottom: 0;
  right: 0;
  background: yellow;
  padding: 2px 5px;
  font-size: 0.75rem;
}
</style>
</head>
<body>
  <div class="box"></div>
</body>
```

- a. What will be the width and height of the div element in pixels.
 - b. What does ::after do?
6. Consider the webpage header layout below and the corresponding HTML. Answer the following questions. (4 marks)



```
<body>
<header class="header">
  <div class="logo">MySite</div>

  <ul class="nav">
    <li><a href="#">Home</a></li>
    <li><a href="#">Services</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Contact</a></li>
  </ul>

  <button class="login-btn">Login</button>
</header>
</body>
```

- a. Write the corresponding CSS file using flexbox for layout.
7. Answer the following. (1+ 3 + 3 marks)
- a. Is Keyword research categorized under on-page SEO or off-page SEO? Justify.
 - b. You consider a keyword phrase for optimization "cheap flights to Goa." Search volume is 50,000, keyword difficulty 70. Should you target this keyword? Would you prefer a short-tail or long-tail alternative keyword phrases? Suggest atleast 2 with justification.
 - c. What is the purpose of sitemap.xml in SEO? Write a simple sitemap.xml file containing 2 URLs (homepage and contact). Differentiate between <priority> and <changefreq> tags in a sitemap.xml with a suitable example.
8. Answer the following. (2+2 marks)
- a. In JSON-LD, explain the role of "@context" and "@type". Why would omitting them break rich snippets?
 - b. If the website fails to include properties such as aggregateRating or datePublished, how does that matter?